

1. Describe the error precisely (nature of problem, location). Fix the problem and submit your corrected source code along with a screenshot of the execution. Develop a robust solution that allows the user to enter any size username (within reason) without massively over allocating memory. (10 points)

The error is in line scanf() when it's assigning the input to the data1 pointer, This input of notARealUsername is 16 bytes big without a terminating character. It is overrunning the boundaries of a malloc() memory allocation.

```
mirambea@eos21:~/cis452/lab9/question1$ ./SampleProgramOneElectricFence

Electric Fence 2.2 Copyright (C) 1987-1999 Bruce Perens <bruce@perens.com>
Please input username:
alec
You entered: [alec]
size of username: 4
mirambea@eos21:~/cis452/lab9/question1$ ./SampleProgramOneElectricFence

Electric Fence 2.2 Copyright (C) 1987-1999 Bruce Perens <bruce@perens.com>
Please input username:
username
You entered: [username]
size of username: 8
mirambea@eos21:~/cis452/lab9/question1$ ./SampleProgramOneElectricFence

Electric Fence 2.2 Copyright (C) 1987-1999 Bruce Perens <bruce@perens.com>
Please input username:
notARealUsername
You entered: [notARealUsername]
size of username: 16
mirambea@eos21:~/cis452/lab9/question1$
```

2. Diagram

Shared C Libraries
Stack: grows downward
Run Time Data (Heap): grows upward
Data
Text
BSS Uninitialized data segment
Initialized Data
Text

3. Run your sample program multiple times. Notice that the dynamic segments appear to "hop around" or re-locate in memory. What's going on here, and why? (5 points)

Address obfuscation is occurring, this is a technique that randomizes the virtual addresses of code and data. It is to help make it harder for attackers to exploit memory errors, as well as to also make it harder to understand the program's structure.

```
ostrocon@eos24:~/CIS452$ gcc -g -Wall lab9ProgramAssignment.c -o l9pa
ostrocon@eos24:~/CIS452$ ./l9pa
Text segment (main function): 0x61d59399c204
Initialized data (global_init): 0x61d59399f010
Uninitialized data (global_uninit): 0x61d59399f018
Heap (malloc'd variable): 0x61d593f786b0
Stack (stack_var): 0x7ffca1bdef4c
Stack address (local_var at depth 3): 0x7ffca1bdef24
Stack address (local_var at depth 2): 0x7ffca1bdeef4
Stack address (local_var at depth 1): 0x7ffca1bdeec4
Shared library (printf function): 0x77c110260100
```